Detailed Explanation of Question 1c
ORF 309 S2025 Midterm 1

1 Problem Setup

Bob takes an exam with 2 questions, each worth 10 points.

Given information:

- $P(A) = 0.8$ (prob of getting question 1 correct)
- $P(B|A) = 0.6$ (prob of getting question 2 correct given question 1 correct)
- $P(B|A^c) = 0.1$ (prob of getting question 2 correct given question 1 wrong)

From parts (a) and (b):

- $P(B) = 0.5$ (total prob of getting question 2 correct)
- $P(A \cap B) = P(B|A)P(A) = 0.6 \times 0.8 = 0.48$

2 Part (c): Find Variance of Bob's Grade

2.1 Define the Random Variable

Let X be Bob's total grade. Using indicator random variables:

$$X = 10 \cdot (1_A + 1_B) \tag{1}$$

where:

- $1_A = \begin{cases} 1 & \text{if Bob gets question 1 correct} \\ 0 & \text{otherwise} \end{cases}$
- $1_B = \begin{cases} 1 & \text{if Bob gets question 2 correct} \\ 0 & \text{otherwise} \end{cases}$

2.2 Step 1: Calculate $E[X]$

By linearity of expectation:

$$\begin{aligned} E[X] &= 10 \cdot E[1_A + 1_B] && (2) \\ &= 10 \cdot (E[1_A] + E[1_B]) && (3) \\ &= 10 \cdot (P(A) + P(B)) && (4) \\ &= 10 \times (0.8 + 0.5) && (5) \\ &= 13 && (6) \end{aligned}$$

Key fact: For any indicator random variable 1_A, we have $E[1_A] = P(A)$.

2.3 Step 2: Calculate $E[X^2]$ — The Critical Step

This is where most students struggle. Let's do it step by step.

$$E[X^2] = E\left[(10(1_A + 1_B))^2\right] = 10^2 \cdot E\left[(1_A + 1_B)^2\right] \tag{7}$$

2.3.1 Expand the Square

Key Algebraic Step

$$(1_A + 1_B)^2 = 1_A^2 + 1_B^2 + 2 \cdot 1_A \cdot 1_B \tag{8}$$

2.3.2 Simplify Using Indicator Properties

For any indicator random variable 1_A:

Important Property

$$1_A^2 = 1_A \tag{9}$$

Why? Because $1_A \in \{0, 1\}$, so $0^2 = 0$ and $1^2 = 1$.

Therefore:

$$(1_A + 1_B)^2 = 1_A + 1_B + 2 \cdot 1_A \cdot 1_B \tag{10}$$

2.3.3 Take Expectation

$$\begin{aligned} E[(1_A + 1_B)^2] &= E[1_A + 1_B + 2 \cdot 1_A \cdot 1_B] && (11) \\ &= E[1_A] + E[1_B] + 2E[1_A \cdot 1_B] && (12) \end{aligned}$$

Key question: What is $E[1_A \cdot 1_B]$?

Product of Indicators

For indicator random variables:

$$1_A \cdot 1_B = 1_{A \cap B} \tag{13}$$

Why? The product equals 1 only when **both** A and B occur.
Therefore:

$$E[1_A \cdot 1_B] = E[1_{A \cap B}] = P(A \cap B) \tag{14}$$

2.3.4 Substitute Values

$$\begin{aligned} E[(1_A + 1_B)^2] &= P(A) + P(B) + 2P(A \cap B) && (15) \\ &= 0.8 + 0.5 + 2 \times 0.48 && (16) \\ &= 1.3 + 0.96 && (17) \\ &= 2.26 && (18) \end{aligned}$$

2.3.5 Final Calculation of $E[X^2]$

$$E[X^2] = 10^2 \cdot E[(1_A + 1_B)^2] = 100 \times 2.26 = 226 \tag{19}$$

2.4 Step 3: Calculate Variance

Using the variance formula:

Variance Formula

$$\text{Var}(X) = E[X^2] - (E[X])^2 \tag{20}$$

$$\begin{aligned} \text{Var}(X) &= 226 - 13^2 & (21)\\ &= 226 - 169 & (22)\\ &= 57 & (23) \end{aligned}$$

Note: The solution PDF has a typo showing $13^2 = 132$, but the correct value is $13^2 = 169$. The final answer of 57 is still correct.

3 Alternative Method: Using Variance of Sum Formula

This method uses the general formula from section 3.2 and is often more intuitive than computing $E[X^2]$ directly.

3.1 Reframe the Problem

Since $X = 10 \cdot (1_A + 1_B)$, we can define:

- $X_1 = 10 \cdot 1_A$ (points from question 1)
- $X_2 = 10 \cdot 1_B$ (points from question 2)
- Then $X = X_1 + X_2$

3.2 Step 1: Find Var(X_1) and Var(X_2)

For an indicator variable 1_A, the variance is:

$$\text{Var}(1_A) = P(A)(1 - P(A)) = 0.8 \times 0.2 = 0.16 \tag{24}$$

Why Does $\mathrm{Var}(1_A) = P(A)(1 - P(A))$?

For any indicator variable 1_A that takes value 1 with probability $P(A)$ and 0 with probability $1 - P(A)$:

Step 1: Calculate $E[1_A]$

$$E[1_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$$

Step 2: Calculate $E[1_A^2]$

$$E[1_A^2] = E[1_A] = P(A) \quad \text{(since } 1_A^2 = 1_A \text{ for indicators)}$$

Step 3: Apply variance formula

$$\begin{aligned}\mathrm{Var}(1_A) &= E[1_A^2] - (E[1_A])^2 \\ &= P(A) - (P(A))^2 \\ &= P(A) - P(A)^2 \\ &= P(A)(1 - P(A))\end{aligned}$$

This is the standard formula for the variance of a Bernoulli random variable.

Similarly:

$$\mathrm{Var}(1_B) = P(B)(1 - P(B)) = 0.5 \times 0.5 = 0.25 \tag{25}$$

Since $X_1 = 10 \cdot 1_A$ and $X_2 = 10 \cdot 1_B$:

$$\mathrm{Var}(X_1) = 10^2 \times \mathrm{Var}(1_A) = 100 \times 0.16 = 16 \tag{26}$$

$$\mathrm{Var}(X_2) = 10^2 \times \mathrm{Var}(1_B) = 100 \times 0.25 = 25 \tag{27}$$

3.3 Step 2: Find Cov(X_1, X_2)

$$\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(10 \cdot 1_A, 10 \cdot 1_B) \tag{28}$$

$$= 100 \times \mathrm{Cov}(1_A, 1_B) \tag{29}$$

From the covariance formula:

$$\mathrm{Cov}(1_A, 1_B) = E[1_A \cdot 1_B] - E[1_A]E[1_B] \tag{30}$$

$$= P(A \cap B) - P(A)P(B) \tag{31}$$

$$= 0.48 - 0.8 \times 0.5 \tag{32}$$

$$= 0.48 - 0.4 \tag{33}$$

$$= 0.08 \tag{34}$$

Therefore:

$$\mathrm{Cov}(X_1, X_2) = 100 \times 0.08 = 8 \tag{35}$$

3.4 Step 3: Apply the Variance of Sum Formula

Variance of Sum Formula

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2) \tag{36}$$

Substituting our values:

$$\begin{aligned}
\text{Var}(X) &= \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2) &(37)\\
&= 16 + 25 + 2(8) &(38)\\
&= 16 + 25 + 16 &(39)\\
&= 57 &(40)
\end{aligned}$$

This matches our previous answer! This method is often cleaner and more intuitive, especially when you need to work with sums of random variables.

4 Key Takeaways

4.1 Why Not Independent?

You might wonder: why can't we use $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$?

Answer: That formula only works when X and Y are **independent**. Here, 1_A and 1_B are **dependent** because:

$$P(B|A) = 0.6 \neq 0.5 = P(B) \tag{41}$$

4.2 General Formula for Variance of Sum

For any two random variables X and Y:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \tag{42}$$

where

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] \tag{43}$$

For our problem:

$$\begin{aligned}
\text{Cov}(1_A, 1_B) &= E[1_A \cdot 1_B] - E[1_A]E[1_B] &(44)\\
&= P(A \cap B) - P(A)P(B) &(45)\\
&= 0.48 - 0.8 \times 0.5 &(46)\\
&= 0.48 - 0.4 &(47)\\
&= 0.08 > 0 &(48)
\end{aligned}$$

The positive covariance indicates that the two questions are positively correlated (doing well on one increases the chance of doing well on the other).

4.3 Summary of Key Facts

1. **Indicator squared equals itself:** $1_A^2 = 1_A$
2. **Product of indicators:** $1_A \cdot 1_B = 1_{A \cap B}$
3. **Expectation of indicator:** $E[1_A] = P(A)$
4. **Expectation of product:** $E[1_A \cdot 1_B] = P(A \cap B)$
5. **Variance formula:** $\mathrm{Var}(X) = E[X^2] - (E[X])^2$
6. **Linearity of expectation:** Always works (no independence needed)
7. **Variance of sum:** Only additive when variables are independent